Let's Date Again

Product Requirements Document

React Native Expo· MVP · Solo Flow · Version 1.0

Platform	Language	Min iOS	Min Android
React Native Expo	**TypeScript**	**14.0**	**API 26**

1. Product Overview

Let's Date Again (LDA) is a relationship wellness app for Indian couples. The MVP is a solo 5-day experience where one user completes a guided daily ritual, each day building emotional context from the last. The app tracks mood, journaling, memories, and personality profile, culminating in a shareable 5-Day Story report.

Core User Journey

5-Day Solo Flow

Onboarding → Day 1 (Spark Check) → Bridge → Day 2 (Mood Room)
Bridge → Day 3 (Assumptions Test) → Bridge → Day 4 (Memory Jar)
Bridge → Day 5 (The Reveal & Report Card)

Design Principles

- No navigation bars on day screens — full-screen immersive flow
- No "Next" buttons — CTAs use emotionally resonant copy
- Playfair Display italic for all key emotional moments
- Haptic feedback at every meaningful interaction
- 5-day progress strip present on all day screens (not bridges/onboarding)
- Each day has a distinct accent colour and emotional theme

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2. Technical Stack & Architecture

Core Dependencies

Concern	Choice
Framework	React Native Expo
Language	TypeScript
Navigation	@react-navigation/native v6 + Stack
State Management	Zustand + Immer middleware
Primary Storage	react-native-mmkv (fast KV store)
Secondary Storage	AsyncStorage (large objects: journal, memories)
Animations	react-native-reanimated v3
Gestures	react-native-gesture-handler v2
SVG / Charts	react-native-svg
Haptics	react-native-haptic-feedback
Image Picker	react-native-image-picker v7
Share / Export	react-native-share + react-native-view-shot
Gradients	react-native-linear-gradient

Architecture Decision: Offline-First

The MVP is 100% offline. All logic runs client-side. JSON data files serve as the content database. The service layer is designed so that every local function can be swapped for an API call without touching screen or store code.

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3. Project Structure

The project follows a feature-first organisation within the src/ directory:

```
src/
  assets/            # Fonts (Playfair Display, Inter), images, Lottie JSONs
  components/
    common/          # ProgressStrip, StreakRing, JarAnimation, MoodCard, etc.
    day1/            # ConnectionSlider, SparkQuiz, PersonalityResult
    day2/            # CandleMoodPicker, MoodFollowUp
    day3/            # AppreciationSnap, AssumptionsTest, MirrorGameResults
    day4/            # MemoryJarInput, TinyCompliment, DailyTwo, DropBox
    day5/            # MoodChart, ReportCard, JarReveal, TheLetter, etc.
  data/              # All static JSON content files
  navigation/        # RootNavigator, DayNavigator, DayRouter logic
  screens/           # One folder per day + bridges + onboarding
  services/          # badgeCalculator, letterGenerator, storage, etc.
  store/             # Zustand stores: useUserStore, useDayStore, etc.
  theme/             # colors, typography, spacing, animations
  utils/             # segmentResolver, personalityCalculator, haptics
```

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4. Data Models (JSON)

4.1 Day Store — Primary Data Model

All 5 days' data is held in a single Zustand store persisted to MMKV. Below is the full TypeScript interface:

Day 1 Data Shape

complete: boolean
completionTimestamp: string | null
sliderScore: number (1–10)
segment: segment_1 | segment_2 | segment_3 | segment_4 | segment_5
quizAnswers: Record<q1–q7, A | B>
personalityType: steady_flame | electric_spark | deep_current | shifting_tide
moodScore: number (= sliderScore, used in Day 5 chart)

Day 2 Data Shape

complete: boolean
intentionWord: string (chosen from 8 options or custom, max 12 chars)
mood: connected | grateful | loved | playful | overwhelmed | frustrated | distant | missed
moodScore: number (3–9, from mood table)

followUpQuestion: string (personalised from D1 Q2 answer)
followUpAnswer: string
followUpLength: number (used in badge Axis A calculation)

Day 3 Data Shape

complete: boolean
intentionWord: string
appreciationSnap: string | null
mirrorAnswers: Record<q1–q10, boolean>
trueRatio: number (0.0–1.0) → used in Axis B & C badge calculation
oneCertainty: string | null
partnerInviteSent: boolean

Day 4 Data Shape

complete: boolean
intentionWord: string
memoryContent: string | null
memoryType: text | photo | emoji | skipped
tinyComplimentWord: string | null
daily2Q1: string (emotion carrying into today)
daily2Q2: string (relationship lesson this week)
daily2Status: both | one | skipped
dropBoxUsed: boolean
dropBoxReframedText: string | null (original never stored)

Day 5 Data Shape

complete: boolean
badgeName: string (one of 8 badges)
badgeTier: gold | standard | emerging
dedicationScore: number (2–8)
promise: string | null
letterGenerated: boolean
averageScore: number (rolling average for report chart)
partnerInviteSent: boolean
soloContinuation: boolean

4.2 Static JSON Content Files

File	Contents	Used By
sparkQuizQuestions.json	5 segments × 7 questions (A/B options each)	D1_SparkQuiz
personalityTypes.json	4 types: name, subLabel, description, traits, growth, triviaFact	D1_PersonalityResult, D4, D5
moodOptions.json	8 moods with scores, candle variant, follow-up questions, axis signals	D2_MoodPicker

File	Contents	Used By
assumptionsSets.json	5 sets × 10 TRUE/FALSE statements, keyed by personality type	D3_AssumptionsTest
badges.json	8 badges with axis keys, descriptions, trait pills	D5_CelebrationBadge
honestMomentCopy.json	5 segment responses with {score} token	D1_HonestMoment
bridgeQuotes.json	Quote per bridge + mood-pattern insight lines for D5 chart	All Bridge screens, D5
intentionWords.json	8 words + metadata	All Bridge screens
triviaFacts.json	4 personality-keyed psychology facts	D4_TriviaFact

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5. Navigation Architecture

React Navigation v6 Stack navigator. No bottom tabs — the app is a linear progressive experience. The DayRouter service resolves which screen to open on every app launch.

Navigation Tree

```
RootNavigator (Stack)
  ├─ OnboardingNavigator → Splash → Welcome → NameInput
  └─ MainNavigator
      ├─ DayRouter (resolves entry screen on launch)
      ├─ Day1Navigator: D1_ConnectionSlider → D1_HonestMoment → D1_SparkQuiz → D1_PersonalityResult
      ├─ Bridge_1to2
      ├─ Day2Navigator: D2_MoodPicker → D2_MoodFollowUp
      ├─ Bridge_2to3
      ├─ Day3Navigator: D3_AppreciationSnap → D3_AssumptionsTest → D3_OneCertainty →
D3_MirrorResults
      ├─ Bridge_3to4
      ├─ Day4Navigator: D4_MemoryJar → D4_TinyCompliment → D4_DailyTwo → D4_TriviaFact →
D4_DropBox
      ├─ Bridge_4to5
      └─ Day5Navigator: D5_Celebration → D5_ReportCard → D5_ThePromise → D5_JarReveal →
D5_TheLetter → D5_PartnerInvite
```

DayRouter Resolution Logic

On every app open, DayRouter checks completion flags in order: D1 → D2 → D3 → D4 → D5. Routes to the bridge/day of the first incomplete day. If 48+ hours have elapsed since last active, checks streak shield eligibility before routing.

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6. Screen Specifications

6.1 Day 1 — The Spark Check

D1_ConnectionSlider

Full-screen, no nav bar. 5-pip progress strip at top (Day 1 active, 2–5 inactive).

- Large animated number (Playfair Display Bold, 96sp) updates live on drag
- Integer-step haptic feedback (Light) at each whole number
- Brand phrase below slider: "no right answer · only your truth" (static, always visible)
- CTA: "That's my number" (not "Next")
- On tap: store sliderScore, compute segment, navigate with number "landing" animation to D1_HonestMoment

D1_HonestMoment

Score displayed large (Playfair, same position as slider number — creates continuity). Personalised copy from honestMomentCopy.json. No auto-advance. Designed for 4–6 second dwell. CTA: "Take the Spark Quiz" with sub-text "7 questions · 90 seconds".

D1_SparkQuiz (7 Questions)

- 7-pip progress bar at top
- Mood badge pill per question (e.g. "Spicy · Self-aware")
- Question in Playfair italic; two choice cards (A/B) below
- Swipe hint disappears after first swipe
- No confirm button — selection auto-advances after 120ms delay
- Unchosen card drops to opacity 0.3; transition: scale compress + slide-up
- After Q7: calculate personality type in background, navigate to D1_PersonalityResult

D1_PersonalityResult

- Animation sequence under 1 second: type name letters fade L→R → icon appears → description slides up → trait pills stagger (80ms each)
- Key line in Playfair italic: "Would your partner say the same thing about themselves?"
- Primary CTA: "Save my result" (triggers shareable card export)
- Ghost CTA: "Come back tomorrow →"
- On either tap OR app close: mark D1 complete, set streak = 1

6.2 Bridge Screens (B1→2, B2→3, B3→4, B4→5)

All bridges share a 4-zone layout. They load on next app open after a day is completed.

Zone	Content	Notes
Zone 1	Streak ring (animated pulse on load)	Shows current streak number
Zone 2	Recap card(s) from previous day	Exact data from last day's key activity
Zone 2b	Intention Word Selector	8 pill options + free-type (max 12 chars). Not on B4→5
Zone 3	Bridge quote in Playfair italic	Pre-written per bridge in bridgeQuotes.json

Zone	Content	Notes
Zone 4	CTA: "Continue to Day N →"	Advances to that day's first screen

Missed Day Logic (Streak Shield)

If ≥48 hours since last active AND streakShieldAvailable = true:
→ Auto-apply shield: show card "Life happened today. Your streak is safe."
→ Set streakShieldAvailable = false for this 7-day cycle
→ Show missed-day bridge variant: "You were away for a day. That's okay."
If shield already used: streak resets to 1 on next active day

Repeat Mood Detection (B3→4)

If same mood selected 3 consecutive days (repeatMoodStreak ≥ 3):
→ Add gentle line after recap: "You've been feeling [mood] for a few days. That's worth noticing. The jar is a good place for that."

6.3 Day 2 — The Mood Room

D2_MoodPicker

Intention word pill shown at top. Centred SVG candle changes flame animation based on mood. 8 mood options. Mood selection triggers candle animation, navigates after 400ms.

- connected: tall warm gold flame
- grateful: steady amber
- frustrated: low blue-tinted flame
- missed: warm but small
- Follow-up question determined by mood + D1 Q2 answer for personalisation

D2_MoodFollowUp

Open text field, auto-growing textarea. Question text from store. On save: store answer, update repeat mood streak, mark Day 2 complete.

6.4 Day 3 — The Assumptions Test

D3_AppreciationSnap (Pre-screen Quick Win)

Prompt: "Before we test what you know about them — what do you want them to know about you right now?" Open text, CTA: "Save this →". On save: card flies into journal with arc animation. Dedication Score +0.5.

D3_AssumptionsTest (10 Questions)

10 TRUE/FALSE statements, full-screen per question. Set determined by D1 personality type. No back. Auto-advances on tap. Progress: "3 of 10" + linear bar.

D3_OneCertainty (Quick Win 2)

Transition screen after mirror game: "What's the one thing you know for certain?" Converts uncertainty from mirror game into warmth. On save: flies into journal. Used verbatim in D5 report.

D3_MirrorResults — Key Conversion Screen

Split screen: left column shows user's 10 answers; right column is completely blurred/frosted with lock icon. The only resolution is inviting their partner.

- Bottom banner: "Since your partner isn't onboarded yet, you can still share this card."
- Share card CTA: exports image via ViewShot + react-native-share
- Invite partner CTA: opens WhatsApp with pre-written message + unique invite link
- Screenshot icon always visible top-right

6.5 Day 4 — The Memory Jar

D4_MemoryJar

SVG jar centred on screen at ~60% fill with 3 paper notes. Three input types: Text · Photo · Emoji + date. On "Drop it in the jar →": note animates as arc into jar, jar shakes, counter updates, fill goes to ~80%.

D4_TinyCompliment (slides up post-drop)

8 word pills: Seen / Safe / Lighter / Lucky / Proud / Loved / Calm / Home + free-type. On tap: sparkle animation on jar, word sealed into jar entry, glowing label inside jar.

D4_DailyTwo

Two open text cards: "What emotion are you carrying into today?" and "What's something your relationship taught you this week?" Saved to journal. Both answers appear in D5 timeline.

D4_TriviaFact

Full-screen, 5 seconds. Personalised psychology fact from personalityTypes.json. Swipe up to dismiss. No CTA. Pure gift moment.

D4_DropBox (Optional)

Ghost link entry. User writes raw feeling. After 20+ chars: "Help me say this better →" appears. 0.8–1.2s deliberate loading pause. Shows original + reframed card. Privacy: original text never stored.

Drop Box Privacy Rules

Original text: NEVER stored, cleared from memory on navigation
Only the final kept/edited reframe is stored (d4_dropBoxReframedText)
drop_box_used: boolean flag stored (for badge calculation)
Rose note appears in D5 jar but content is NOT shown to preserve privacy

6.6 Day 5 — The Reveal

D5_CelebrationBadge

ALL 5 PIPS FILLED — fill animation L→R over 600ms. Confetti (2 seconds, all 5 day accent colours). Badge calculated from full axis scoring. Key lines: "That's not nothing." and "Most people quit at Day 2."

D5_ReportCard (4 Sections)

- Section 1: 5-bar mood chart with pre-written insight line (4 pattern variants)
- Section 2: Personality type deep-dive — 3 strengths + 1 growth area
- Section 3: Week in Moments — vertical journal timeline, exact user words, intention word pills
- Section 4: Memory jar preview (full reveal on next screen)

D5_ThePromise

"Five days in. One thing you want to do differently from tomorrow." Saved as closing statement of report card. Optional inclusion on shareable card.

D5_JarReveal

Jar fills from 80% to 100% over 1.2s. Golden glow pulses. All 5 day-coloured notes with 80ms stagger bounce. Notes tappable for memory content. Tiny compliment word glowing inside.

D5_TheLetter

Template-generated personal letter using: user name, D1 slider score, personality type name, first 8 words of D4 memory. No buttons for 6 seconds. Then: "Save this" + "Share this".

D5_PartnerInvite

"What do you want to do next?" Primary: "Invite your partner" → WhatsApp invite. Ghost: "Continue solo for now" → Daily 2 becomes ongoing habit from Day 6+.

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7. State Management

7.1 Zustand Stores

Store	Persisted	Key Contents
useUserStore	MMKV	userId (UUID), name, onboardingComplete, createdAt
useDayStore	MMKV	Full DayData (all 5 days), all flags, quiz answers, text content
useStreakStore	MMKV	streakCount, shieldAvailable, shieldUsed, lastActiveDate, repeatMoodStreak
useJournalStore	AsyncStorage	JournalEntry[], JarMemory[], jarFillLevel
useSessionStore	None (in-memory)	currentDay, currentScreen, sessionStartTime

7.2 Badge Calculator — Axis Scoring

Runs on D5 completion using all stored data. Three axes each independently resolve to one of two values, then mapped to one of 8 badges.

Axis A: Expressive vs Active

Score EXPRESSIVE (+1 each): D1 Q1=A, D1 Q6=A, D2 followUpLength ≥60 chars, D4 Daily2 both ≥40 chars, D4 memory=text
Score ACTIVE (+1 each): D1 Q1=B, D1 Q6=B, D4 memory=photo (+2), D2 mood=playful or connected
Resolve: ≥3 signals wins. Tiebreak: D1 Q1 answer decides.

Axis B: Deep vs Present

Score DEEP (+1 each): D1 Q4=B, D1 Q7=B, D2 mood=loved/grateful/missed, D3 trueRatio ≥0.7 (+2), dropBoxUsed (+2)
Score PRESENT (+1 each): D1 Q4=A, D1 Q7=A, D2 mood=connected/playful/overwhelmed, D3 trueRatio ≤0.4 (+2)
Resolve: ≥3 signals wins. Tiebreak: D1 Q7 answer decides.

Axis C: Protecting vs Building

Score PROTECTING (+1 each): D1 slider ≥6, D1 Q5=B, D1 Q6=A, D3 trueRatio ≥0.7, D2 mood=grateful/loved
Score BUILDING (+1 each): D1 slider ≤5, D1 Q5=A, D1 Q7=A, dropBoxUsed, D2 mood=frustrated/distant/overwhelmed
Resolve: ≥3 signals wins. Tiebreak: slider ≥6 = Protecting; ≤5 = Building.

Badge Mapping (Axis A × B × C)

Expressive + Deep + Protecting → The Warm Keeper
Expressive + Deep + Building → The Honest Architect
Active + Present + Protecting → The Joyful Anchor
Active + Present + Building → The Curious Lover
Active + Deep + Protecting → The Quiet Strength
Active + Deep + Building → The Steady Climber
Expressive + Present + Protecting → The Spark Keeper
Expressive + Present + Building → The Spark Keeper
All axes tied → The Intentional Partner

Dedication Score → Badge Tier

Base score for completing all 5 days: 4.0 (guaranteed)
+0.5 for each Quick Win completed (D2 followup, D3 Appreciation Snap, D3 One Certainty, D4 Tiny Compliment, D4 Daily Two, D4 Drop Box, D5 Promise)
6–8 total → Gold tier (gold ring, full confetti, shareable card)
4–6 total → Standard tier
2–4 total → Emerging tier (warmer framing)

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8. Animations & Micro-Interactions

All animations use react-native-reanimated v3. Use useNativeDriver via worklets. Prefer withSpring for physical feel, withTiming for controlled transitions.

Animation	Implementation	Duration
Slider number live update	withSpring on value, Playfair 96sp	~200ms
Integer haptic on slider drag	react-native-haptic-feedback (Light)	Instant
Number "landing" D1-1→D1-2	Shared value interpolation (mock shared element)	400ms
Quiz card auto-advance	withTiming opacity 0.3 + translateY slide-up	120ms delay + 200ms
Personality type letter reveal	withDelay stagger per character, withFadeIn	~600ms total
Streak ring pulse on bridge load	withSequence(withSpring(1.15), withSpring(1))	300ms
Memory note fly into jar	Interpolated arc path + opacity + scale	500ms
Jar fill level change	withTiming height to target	800ms
Jar shake after drop	withSequence translateX oscillation (5 steps)	400ms
Tiny compliment sparkle	Lottie sparkle animation on jar	400ms
D5 pip fill L→R	withDelay per pip + withTiming fill colour	600ms total
D5 confetti	Lottie confetti (5 day accent colours)	2000ms
Badge reveal sequence	Name letters fade + icon scale + pills stagger 80ms each	<800ms
D5 jar 100% fill + glow	withTiming height + withRepeat(1) opacity pulse	1200ms
Note bounce in jar	withSpring scale with 80ms stagger per note	~500ms
Card flies to journal	Animated arc translateX/Y + opacity	500ms

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9. Local Storage Strategy

9.1 What Is Stored Where

Data	Store	Key / Notes
User profile (name, userId)	MMKV (Zustand)	lda-user-store
All 5 days data + flags	MMKV (Zustand)	lda-day-store
Streak state + shield	MMKV (Zustand)	lda-streak-store
Journal entries	AsyncStorage	lda-journal-entries (larger objects)
Jar memories	AsyncStorage	lda-jar-memories
Shareable card image	Temp file (cache dir)	Cleared after share / session end
Partner invite link	MMKV	lda-invite-link

Data	Store	Key / Notes
Drop box original text	NEVER STORED	Cleared from memory on navigate

9.2 Data Lifecycle

- All data is local-only in MVP. Nothing sent to any server.
- User can clear all data from Settings screen (with confirmation).
- Day 6+: Journal entries persist. New week cycle begins with fresh DayData (badge + personality type preserved in user profile).
- MMKV reads/writes complete in < 1ms. No UI blocking on storage operations.

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10. Feature Flags & Key Logic

10.1 Shareable Card Generation

Uses react-native-view-shot to capture a hidden off-screen component. Shared via react-native-share.

What APPEARS on shareable card

App name (small watermark), personality type name, badge name + tier
Mini 5-bar mood chart, score line: "Started at [N] · Ended at [N]"
Tiny compliment word as pill, optional Promise text (user toggles)

What NEVER appears on shareable card

Any journal text (D2 follow-up, D4 Daily 2)
Mirror Game answers (D3)
Drop Box content (D4)
Appreciation Snap or One Certainty text (D3)
Only scores, type/badge names, compliment word, and user-opted Promise

10.2 Local Notifications

Day	Title	Body
Day 2	Your jar is waiting.	Day 2 of 5. 5 minutes is all it takes.
Day 3	The Assumptions Test is ready.	How well do you actually know them?
Day 4	Drop a memory today.	The jar needs one more.
Day 5	Day 5. The one that matters.	Everything you built this week becomes visible today.

Scheduled at user's preferred time (default 8:00 PM). Cancelled if day is already completed. Permission requested after D1 completion.

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11. Non-Functional Requirements

Requirement	Target
Cold start time	< 2 seconds on mid-range Android (Snapdragon 665)
Animation frame rate	≥60 fps for all transitions (useNativeDriver via Reanimated worklets)
MMKV read/write	< 1ms
Touch target size	Minimum 44×44pt for all interactive elements
Image storage	Compress to max 2MB before reference saved
Offline operation	100% functional offline — WhatsApp invite is the only network operation
Accessibility	accessibilityLabel on all icon-only buttons; Dynamic Type support
Reduced motion	Detect via AccessibilityInfo.isReduceMotionEnabled(); skip non-essential animations
Privacy	No analytics, no crash reporting in MVP. No data leaves device.

Typography Setup

Two font families required. Both must be linked for iOS (info.plist UIAppFonts) and Android (assets/fonts/):

- Playfair Display: Regular, Italic, Bold, BoldItalic — for all emotional/headline moments
- Inter: Regular, Medium, SemiBold — for all UI, labels, and body copy

Day Accent Colour System

Day	Theme	Accent Hex
Day 1	Bold & Fun	#E85C7A (Coral Pink)
Day 2	Calm & Warm	#9B8EC4 (Lavender)
Day 3	Reflective	#4FBFA5 (Mint Green)
Day 4	Warm & Emotional	#F0A057 (Amber)
Day 5	Bold & Emotional	#D4A843 (Gold)

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12. API Replacement Plan

Every local data function is wrapped in a service layer (src/services/). The JSON and API implementations share identical TypeScript interfaces. When the backend is ready, replace only the service implementation — screens and stores require zero changes.

Current (JSON/Local)	Future (API Endpoint)	Service File
sparkQuizQuestions.json	GET /content/quiz-questions?segment={id}	contentService.ts
personalityTypes.json	GET /content/personality-types	contentService.ts
badgeCalculator.ts (client)	POST /compute/badge	badgeCalculator.ts

Current (JSON/Local)	Future (API Endpoint)	Service File
letterGenerator.ts (template)	POST /generate/letter	letterGenerator.ts
toneReframer.ts (rule-based)	POST /ai/reframe	toneReframer.ts
AsyncStorage journal	POST /journal/entry + GET /journal/{userId}	journalService.ts
Local UUID invite link	POST /invite/generate → short link	inviteService.ts
Mood score local table	Returned from POST /session/mood	moodService.ts

Migration Pattern

Each service file exports async functions with identical signatures in both versions.
Example: getQuizQuestions(segment) returns the same shape whether sourced from JSON or API.
Replace the single implementation line inside the service function to switch from local to remote.
No screen file, no store action, no component requires any change on migration.

For partner flow PRD, see LDA_PRD_Partner_v1 (future document)